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COS30045 Data Visualisation

Class 1

Agenda

- Overview of:
 - Lab Tasks
 - Assessment
 - Demonstrations
- Lab Exercise 0: Set up tasks
- Lab Exercise 1: Data processing



Time goes faster and you have more fun with friends!

- introduce your self to someone you don't know while waiting for class to start!

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Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

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Labs (aka Classes)

- Work on Lab Exercises
- Do demonstrations
- Work on Data Visualisation Project
- Get advice and help from tutor
- No help desk hours for this unit
- Make appointment with tutor to see outside of class time



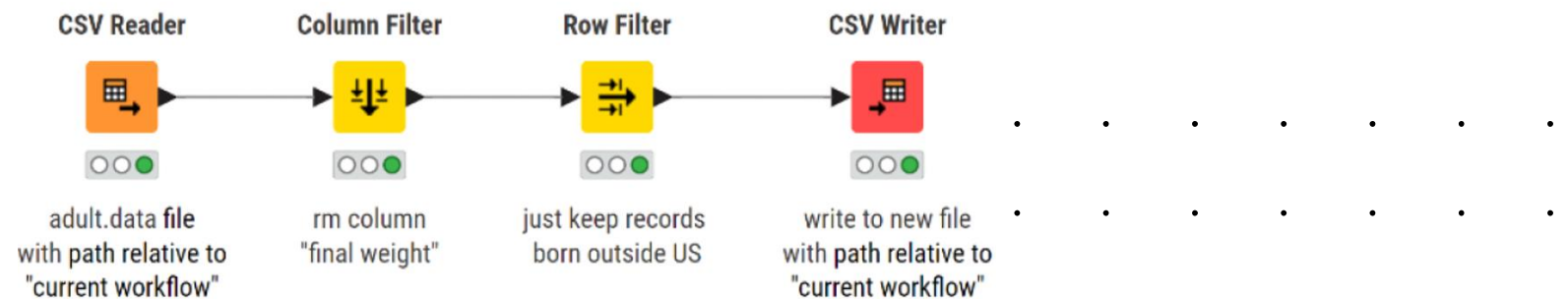
```
</body>
```

```
<!-- Load your scripts here -->
```

```
<script src="https://d3js.org/d3.v7.min.js"></script>
```

```
<script src="js/main.js"></script>
```

```
</body>
```



Assessment

Data Visualisation Assessments

Lab Demonstrations (individual)	15%	Data processing, exploration, visualisation
Assignment (individual)	10%	Data visualisation critique
Project (team and individual)	60%	Visualisation design, website, reflection
	15	Stand-Ups

Exercise Demonstrations (15%)

Assessment

- Submit required files to Canvas **night before** Class
- Attend **On-Campus** Demonstration

Resubmissions: Not permitted

Late penalties: Late demonstration is not accepted without an extension

Demonstration 1 – Week 4!

Demonstration 2 – Week 7!

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Exercise Demonstrations (15%)

You are allowed to use GenAI to help you **prepare** your work.

However, you must be able to:

- Explain works submitted (e.g., data processing logic, design decisions, what code does etc) without referring to comments in code
- Explain the process you used to arrive at a solution
- Make changes to work (e.g., present a different view of data, change code to accept different data format etc)

All code must:

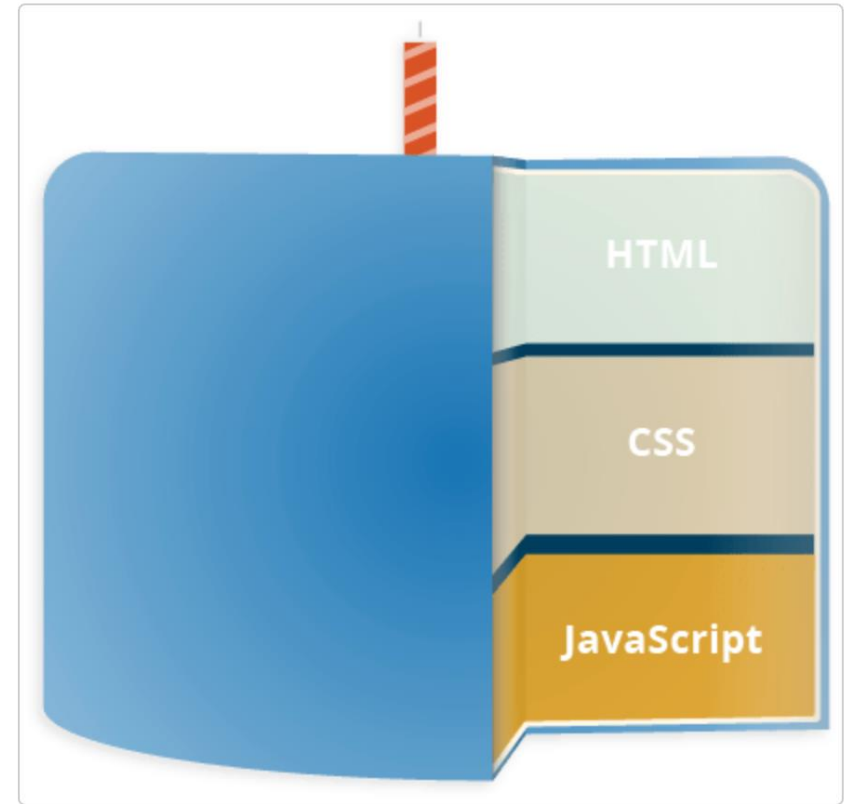
- Be stored on COS30045 GitHub site and demonstrate regular commit history
- Identify use of GenAI code

IMPORTANT: Excessive commenting of code, or heavy reliance on comments in code during a demonstration, will be deemed to be 'use of GenAI'.

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Exercise 0.0: Set up (refer to Canvas)

- Exercise 0.1 Set up GitHub repo on COS30045 GitHub Classroom
- Exercise 0.2 Create test website that demonstrates:
 - HTML
 - CSS
 - JavaScript
- Exercise 0.3 Host website on Mercury



Arrange for consultation with tutor **this week** if you are having difficulty with Week 0 tasks.

Exercise 1.0 Data Processing with KNIME (follow Canvas guide)

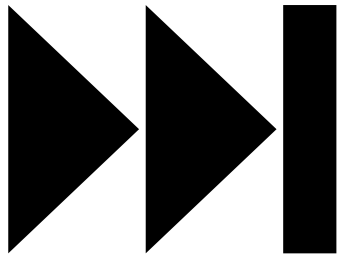
Scenario

Imagine you have been asked to create a set of visualisations that allows users to explore the energy consumption of televisions currently available on the Australian market.

Overview

1. Find the data set
2. Open data set in KNIME
3. Become familiar with data set
4. Clean the data set
5. Transform the data set





Next Week

Exercise 2: Data Exploration and Chart Types in KNIME

- Become familiar with more data processing concepts and KNIME



Open for Innovation

KNIME

